

Cultural Analytics of Gender Representation in Film: A James Bond Case Study

Jakob Lambert-Hartmann^{a,*}

^aUniversität Leipzig

*Corresponding author. E-mail address: wt76geru@studserv.uni-leipzig.de

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1. Introduction

Media has a long history of portraying gender in their stereotypical gender roles. Especially older blockbuster films such as Indiana Jones [1] portray the man (Dr. Jones) as fearless “tough guy” who gets what he wants. Women on the other hand are portrayed as love interest, hysterical, and afraid. This portrayal seems out of date, and films today, albeit still not perfect in their gender representation, would be unlikely to release with such messages. But when did this change occur? Was the change gradual, or did some events/movements bring forward bigger leaps of change?

In this document we investigate how the portrayal of women has changed in the last decades. Specifically, we have chosen the James Bond film series as subject of research. To make our findings quantifiable, the films were analyzed using pretrained AI models, notably YOLO [2] and CLIP [3]. Given the computer guided approach, we also investigate whether this method yields valid and meaningful results.

In the process we have created an analysis framework, which can be used for other research questions. This resulting framework is corpus agnostic, and can be adapted to detect different personality traits. It is open source and free to use [4].

2. Research Question

In this document we question whether women in James Bond are depicted in stereotypical gender roles, and whether this depiction has changed over the years. Based on our observation from Indiana Jones, we have chosen to measure two traits: Sexualization and Agency (describing whether a person is in control of the situation, or whether the person is helpless).

3. Related Works

Neuendorf (2009) [5] analyzed the portrayal of women in the first 20 James Bond films. A trend was discovered that women in later movies were subjected to more violence and were involved in more sexual activities. However the authors also pointed out that the roles of women changed from very limited roles to more autonomous and active roles. In our study we hope eliminate the factor of the changing roles, as our analysis focuses on the appearance rather than the activity.

4. Dataset

The corpus chosen for our analysis is the James Bond 50th Anniversary DVD Collection. It was chosen for several reasons: **Long Running Series:** The collection contains films starting from 1962 to 2008 allowing for an observation period of 46 years. **Series Containing Many Films:** The Collection contains 22 films. The release interval varies from film to movie, but overall they were released mostly regularly. The longest time between releases was 6 years, but the average is around 2 years. **Limiting Confounding Factors:** By limiting our observation to one series, we limit confounding factors. James Bond films all are of the same genre, mostly consist of the same slowly changing cast and crew, and follow similar plots. This makes comparison easier than comparing films of completely different genres (such as Indiana Jones, and Barbie). **Notoriety For Gender Representation:** The James Bond series is notorious for its representation of women, even coining the term Bond girl [6]. The prior research on this Series also allows us to validate our methodology, to test whether it can be applied to other corpora. The choice of only using James Bond movies however also limits the generalizability to other films.

The films were digitized using VLC. Due to copyright restrictions, they are not provided with the code-base. The code-base however is corpus agnostic and can be applied to any other collection of films.

5. Methods

To retrieve the portrayal of women from the raw film data several steps were required. First, the individual films were split into their scenes and saved as images. Then, object detection was performed to get bounding boxes of persons in the scene images. Finally, trait and gender detection was performed on the detected bounding boxes.

5.1. Scene detection

First, the video data was split into scene data. We used PySceneDetect [7] to automate this process. For every scene detected by PySceneDetect, one frame in the middle of the scene was saved as image. This resulted in 800-2000 frames per film totaling 31000 frames.

5.2. Person detection

The next step applied person detection on every scene frame. This was performed with YOLO [2]. The pretrained yolo11 model was used, limited to only person detection. Furthermore the minimum confidence was set to 0.5. This was used in order to limit person detection to people in the foreground.

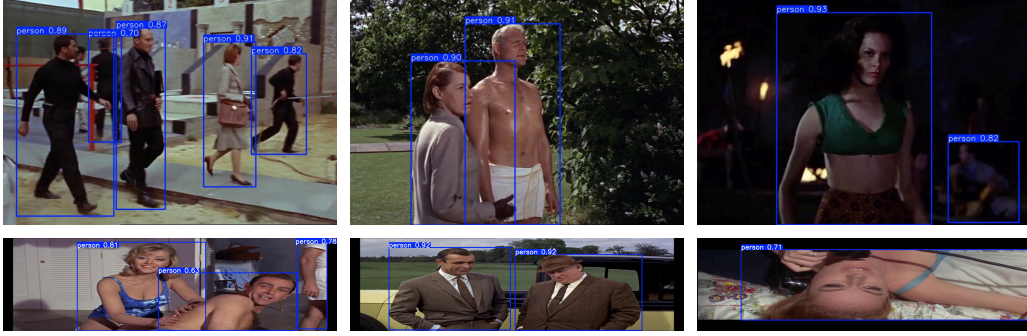


Figure 1: persons detected with YOLO11

5.3. Trait and Gender Detection

The last step in the pipeline performs trait and gender detection for each person bounding box, which YOLO detected. Trait and Gender detection are both performed using CLIP [3]. **Images** used are cropped along the bounding boxes to only contain the detected person and the images were processed in a batch size of 100 images. **Text** prompts were used to determine gender, and prompt ensembles were used for traits to improve performance [8].

trait	negative	positive
gender	“a photo of a woman”	“a photo of a man”
agency	“a person overwhelmed with the situation”	“a person in control of the situation”
	“a person who is passive”	“a person who gets what they want”
	“a submissive person”	“a dominant person”
sexualization	“a sexualized person”	“a professional person”
	“a person in a swimsuit”	“a person in formal attire”
	“a scantily clad person”	“an adequately dressed person”

The following formula used to transform logits to a score from 0 to 1, indicating how strongly the negative traits apply. To determine the degree of sexualization for instance, the mean would be calculated from the logits a corresponding to one ensemble.

$$\text{score} = \sigma\left(\overline{\text{logits}_{\text{trait}}^-}, \overline{\text{logits}_{\text{trait}}^+}\right) \quad (1a)$$

$$\overrightarrow{a_-} = \begin{pmatrix} a_{\text{"sexualized"}} \\ a_{\text{"swimsuit"}} \\ a_{\text{"scantily clad"}} \end{pmatrix} \quad (1b)$$

$$\overrightarrow{a_+} = \begin{pmatrix} a_{\text{"professional"}} \\ a_{\text{"formal attire"}} \\ a_{\text{"adequately dressed"}} \end{pmatrix} \quad (1c)$$

$$\text{score} = \sigma(\overrightarrow{a_+}, \overrightarrow{a_-}) \quad (1d)$$

5.3.1. Limitations

This approach has limitations as can be seen in Fig. 3, where traits of a partial person are analyzed, and in Fig. 5, where a person in the background impacts the detected traits. Additionally, the model and the prompt bias can also contribute to a difference between genders. Finally, the chosen prompt ensembles describe vague concepts such as “sexualized” or “dominant”, which may be difficult for a model to understand. A change to concrete prompts may increase the performance. However, by averaging over multiple such terms, we hope that we can still achieve meaningful results. We will investigate the validity of the prompts later in Section 6

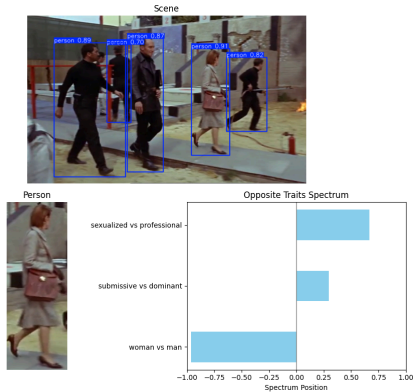


Figure 2: The model correctly recognizes five persons in the image. It recognizes the woman and interprets her as dominant and professional. This aligns with the film, as the woman is the leader of the depicted group



Figure 3: The gender is correctly inferred, but the degree of sexualization is clearly wrong. It seems the naked man in the background contributes to this score, even though he is not the person of interest

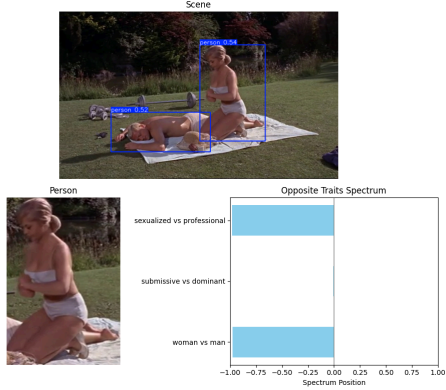


Figure 4: The model recognizes two persons in the image. It recognizes the woman as woman and that her depiction is sexualized

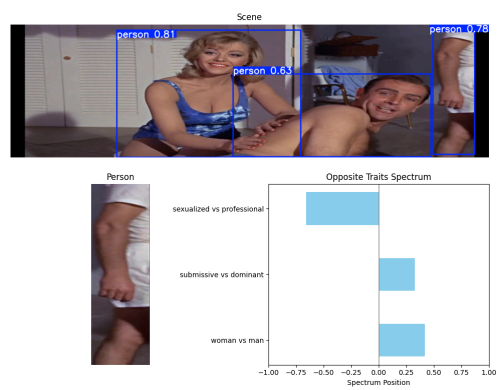


Figure 5: While the gender is correctly inferred, only the arm is visible. The model still predicts the degree of sexualization and dominance.

6. Findings

We visualized the agency score by gender and film and fitted a linear regression model to the data (see Fig. 7). We found that, in contrast to the representation of women in Indiana Jones, the models predicted that women in James Bond were not significantly more helpless than men. This fits the subjective observation we have made, that women in James Bond are commonly portrayed as confident and seductive.

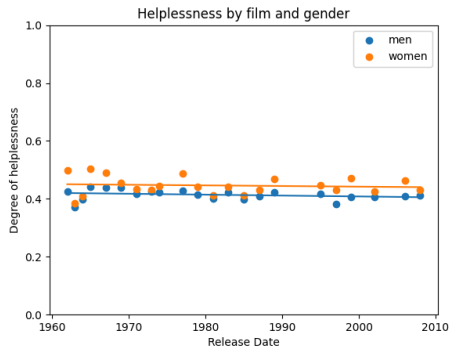


Figure 7: degree of helplessness in films by gender. Observed data and fitted linear regression model

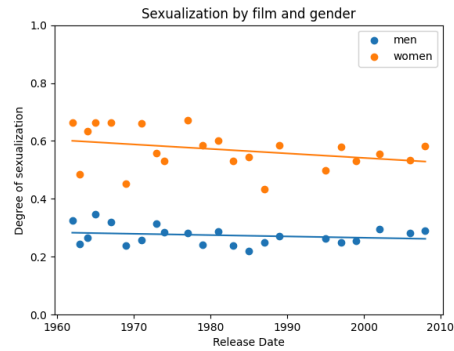
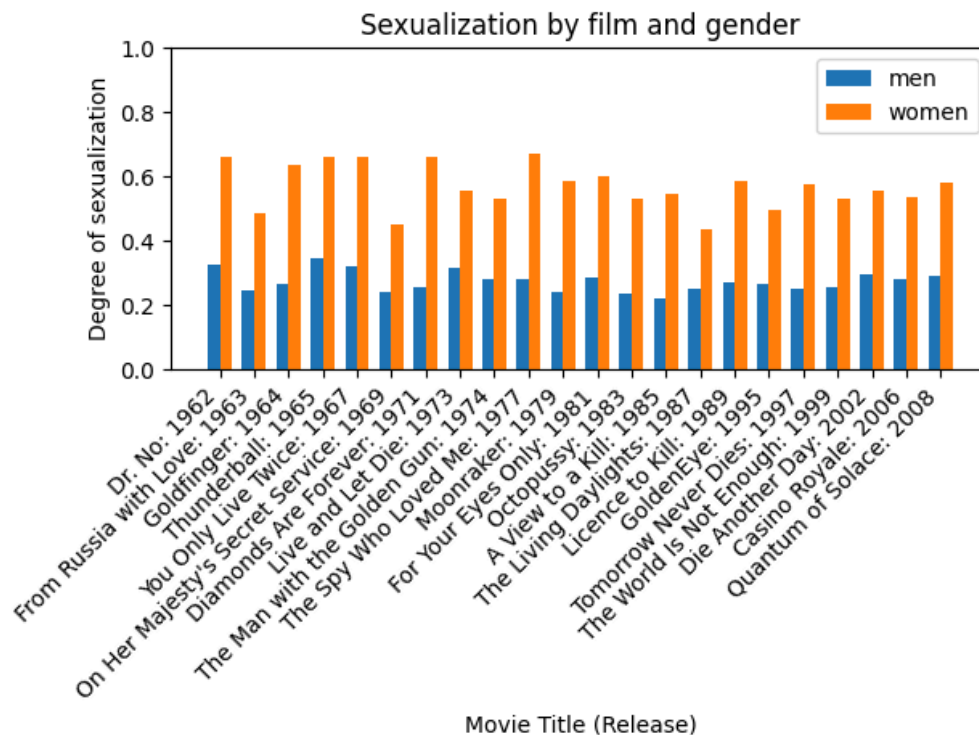
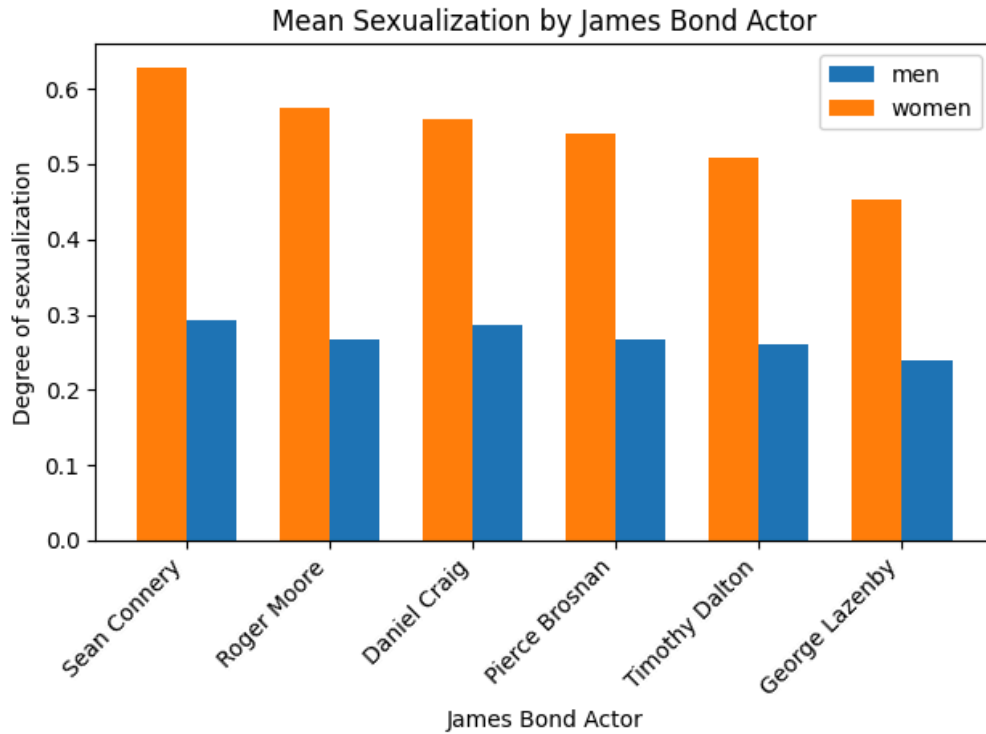


Figure 8: degree of sexualization in films by gender. Observed data and fitted linear regression model

For the sexualized representation however, there was a stark difference between gender (see Fig. 8). The mean sexualization score in men is 0.27, whereas in women it is 0.56. The Linear regression model shows a trend toward less sexualization both in men and women. Where the sexualization scores for men and women in 1962 were at 0.28 and 0.60, in 2008 those scores dropped to 0.26 and 0.53.



To validate our methods we plotted sexualization scores by titles where some outliers became apparent, notably *From Russia with Love: 1963* and *On Her Majesty's Secret Service: 1969*. Those outliers can be well explained by the film plots. In *From Russia with Love* James Bond is accompanied on his mission by a female spy whereas in other films most women are only acquaintances he sleeps with. Thus, there is more screen-time to normalize the sexualization score of women. In *On Her Majesty's Secret Service* James Bond marries a woman. The film is thus more romantic and less sexualized than others in the series



To further validate our methods we investigated the mean sexualization score by James Bond Actor [9]. Daniel Craig is deemed to be the least sexualized James Bond authors, however with our methods this could not be validated. The corpus used for this observation however only contained two films (*Casino Royale* and *Quantum of Solace*), missing *Skyfall*, *Spectre*, and *No Time to Die*. Including those missing titles into the analysis might yield different results.

7. Conclusion

In conclusion, we found that there is a slight trend towards less sexualization in the James Bond film Series. This trend exists both in men and women, but is weaker in men. We could not observe sudden changes, which could have been

expected during waves of feminism for instance [10]. Instead the change was gradual and slow.

Outliers helped in validating our methods, however the strongly varying score also highlights, that the limited sample size of 22 is not sufficient to generalize from the portrayal of women in James Bond to the overall portrayal of women in film. Further studies with larger corpora are warranted in order to monitor general trends. Other long-running film series would be well suited to limit confounding factors.

References

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